

4 Carboxylic acid derivatives

- a demonstrate an understanding that these include acyl chlorides and esters and recognise their respective functional groups, giving examples of molecules containing these functional groups
- b describe and carry out, where appropriate, the reactions of acyl chlorides limited to their reaction with:
 - i water
 - ii alcohols
 - iii concentrated ammonia
 - iv amines
- c describe and carry out, where appropriate, the reactions of esters. This will be limited to:
 - i their hydrolysis with an acid
 - ii their hydrolysis with a base, eg to form soaps
 - iii their reaction with alcohols and acids to explain the process of trans-esterification and recall how it is applied to the manufacture of bio-diesel (as a potentially greener fuel) and low-fat spreads (replacing the hydrogenation of vegetable oils to produce margarine)
- d demonstrate an understanding of the importance of the formation of polyesters and describe their formation by condensation polymerization of ethane-1,2-diol and benzene-1,4-dicarboxylic acid.

4.9 Spectroscopy and chromatography

Knowledge of the concepts introduced in *Unit 2, Topic 2.12: Mass Spectra and IR* will be assumed and extended in this topic.

Students will be assessed on their ability to:

- a explain the effect of different types of radiation on molecules and how the principles of this are used in chemical analysis and in reactions, limited to:
 - i infrared in analysis
 - ii microwaves for heating
 - iii radio waves in nmr
 - iv ultraviolet in initiation of reactions
- b explain the use of high resolution nmr spectra to identify the structure of a molecule:
 - i based on the different types of proton present from chemical shift values
 - ii by using the spin-spin coupling pattern to identify the number of protons adjacent to a given proton
 - iii the effect of radio waves on proton spin in nmr, limited to ^1H nuclei
 - iv the use of magnetic resonance imaging as a non-invasive technique, eg scanning for brain disorders, or the use of nmr to check the purity of a compound in the pharmaceutical industry
- c demonstrate an understanding of the use of IR spectra to follow the progress of a reaction involving change of functional groups, eg in the chemical industry to determine the extent of the reaction
- d interpret simple mass spectra to suggest possible structures of a simple compound from the m/e of the molecular ion and fragmentation patterns
- e describe the principles of gas chromatography and HPLC as used as methods of separation of mixtures, prior to further analysis (theory of R_f values not required), and also to determine if substances are present in industrial chemical processes.